

Stress Analysis Report



Analyzed File:	Combustion System Compl Assembly - New.iam
Autodesk Inventor Version:	2020.1 (Build 241239000, 239)
Creation Date:	5/28/2026, 8:58 PM
Study Author:	Eng Abdul Rahman Abu Zeid
Summary:	

Project Info (iProperties)

Summary

Author	Eng Abdul Rahman Abu Zeid
--------	---------------------------

Project

Part Number	Combustion System Compl Assembly - New
Stock Number	1
Description	Assembly Plenum Chamber 4.25" - A - V1
Designer	Eng Abdul Rahman Abu Zeid
Cost	\$0.00
Date Created	5/28/2026

Status

Design Status	WorkInProgress
---------------	----------------

Physical

Mass	322.225 kg
Area	5991960 mm^2
Volume	40278200 mm^3
Center of Gravity	x=-187.347 mm y=-0.610008 mm z=0.039413 mm

Static Analysis:1

General objective and settings:

Design Objective	Single Point
Study Type	Static Analysis
Last Modification Date	5/28/2026, 7:11 PM
Detect and Eliminate Rigid Body Modes	No
Separate Stresses Across Contact Surfaces	No
Motion Loads Analysis	No

Mesh settings:

Avg. Element Size (fraction of model diameter)	0.1
Min. Element Size (fraction of avg. size)	0.2
Grading Factor	1.5
Max. Turn Angle	60 deg
Create Curved Mesh Elements	No
Use part based measure for Assembly mesh	Yes

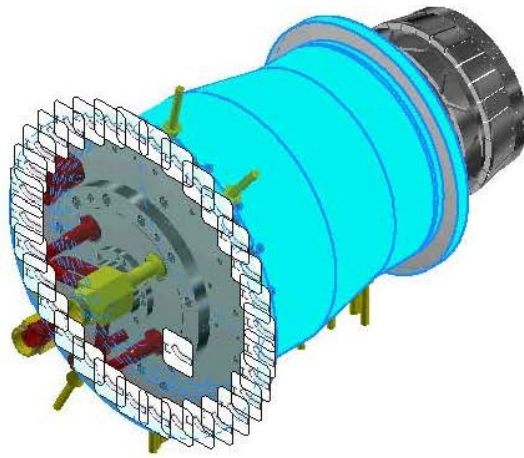
Material(s)

Name	Stainless Steel	
General	Mass Density	8 g/cm ³
	Yield Strength	250 MPa
	Ultimate Tensile Strength	540 MPa
Stress	Young's Modulus	193 GPa
	Poisson's Ratio	0.3 ul
	Shear Modulus	74.2308 GPa
Part Name(s)	Part # 3 - Combustion System Compl.ipt Part # 2 - Combustion System Compl.ipt Part # 1 - Combustion System Compl.ipt Part # 4 - Combustion System Compl.ipt Part # 5 - Combustion System Compl.ipt Part # 6 - Combustion System Compl.ipt Shaft.ipt Tube # 1.ipt Tube # 1.ipt Bolt # 1.ipt Tube # 3.ipt Tube # 4.ipt Tube # 5.ipt Tube # 6.ipt	
Name	Stainless Steel 316	
General	Mass Density	8 g/cm ³
	Yield Strength	250 MPa
	Ultimate Tensile Strength	540 MPa
Stress	Young's Modulus	193 GPa
	Poisson's Ratio	0.3 ul
	Shear Modulus	74.2308 GPa
Part Name(s)	Tube # 2.ipt	

Operating conditions**Force:1**

Load Type	Force
Magnitude	100000.000 N
Vector X	-100000.000 N
Vector Y	-0.000 N
Vector Z	0.000 N

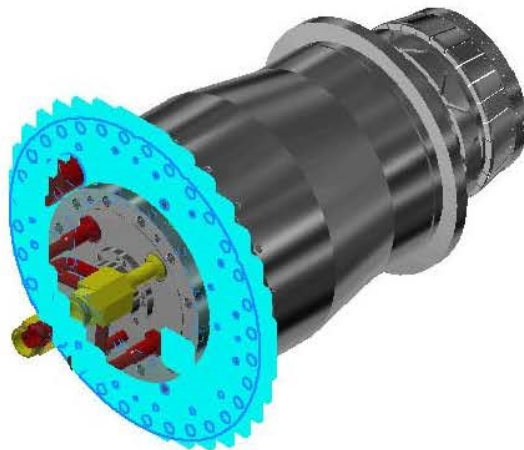
Selected Face(s)



Fixed Constraint:1

Constraint Type	Fixed Constraint
-----------------	------------------

 Selected Face(s)



Results

Reaction Force and Moment on Constraints

Constraint Name	Reaction Force		Reaction Moment	
	Magnitude	Component (X,Y,Z)	Magnitude	Component (X,Y,Z)
Fixed Constraint:1	6616.52 N	6613.77 N	19.0787 N m	0 N m
		156.216 N		11.3382 N m
		109.428 N		15.3442 N m

Result Summary

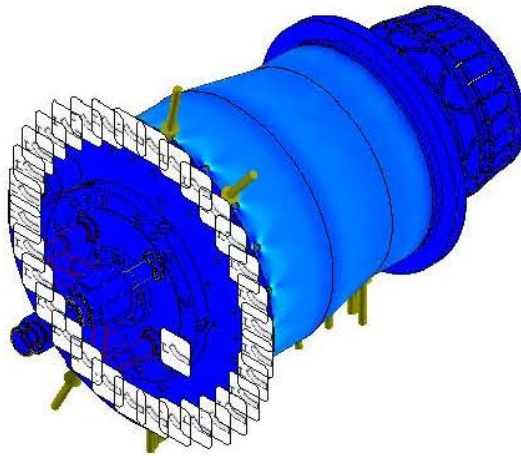
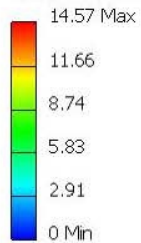
Name	Minimum	Maximum
Volume	40278100 mm ³	
Mass	322.225 kg	

Von Mises Stress	0.000000000350255 MPa	14.5715 MPa
Displacement	0 mm	0.00772403 mm
Safety Factor	15 ul	15 ul
Equivalent Strain	0.00000000000000157284 ul	0.0000670255 ul
1st Principal Strain	-0.000000514469 ul	0.0000241295 ul
3rd Principal Strain	-0.0000757002 ul	0.000000150068 ul

Figures

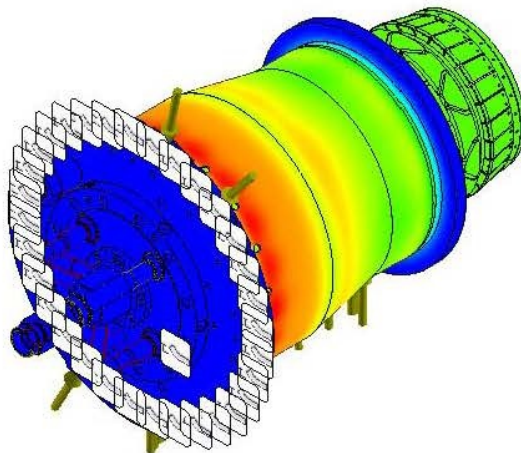
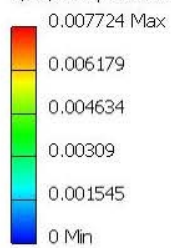
Von Mises Stress

Type: Von Mises Stress
Unit: MPa
5/28/2026, 8:58:02 PM



Displacement

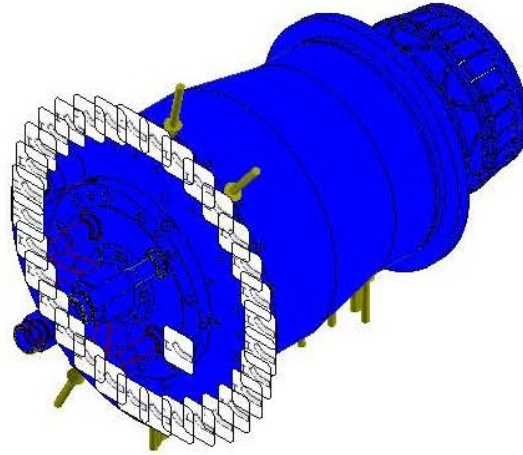
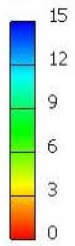
Type: Displacement
Unit: mm
5/28/2026, 8:58:07 PM



Safety Factor

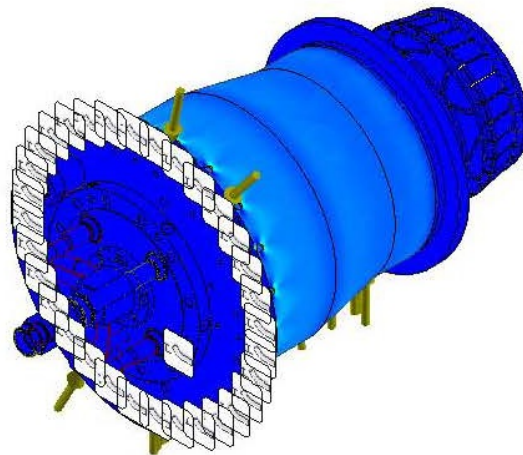
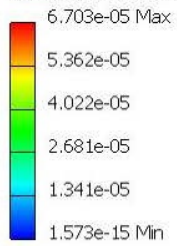
☐ Safety Factor

Type: Safety Factor
Unit: ul
5/28/2026, 8:58:05 PM



☐ Equivalent Strain

Type: Equivalent Strain
Unit: ul
5/28/2026, 8:58:10 PM

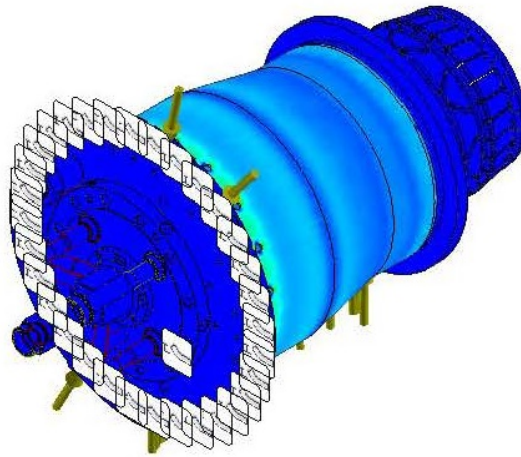
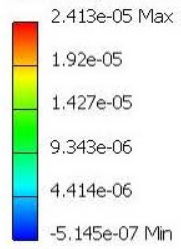


1st Principal Strain

Type: 1st Principal Strain

Unit: ul

5/28/2026, 8:58:12 PM



3rd Principal Strain

Type: 3rd Principal Strain

Unit: ul

5/28/2026, 8:58:14 PM

